Vel Tech Rangarajan Dr. Sagunthala R&D Institute of Science and Technology

School of Computing

Department of Computer Science and Engineering

10211CS224 / FULL STACK APPLICATION DEVELOPMENT (JAVA)

PROJECT REVIEW : MILESTONE PROJECT – 1

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SUBSCRIPTION-BASED MEMBERSHIP ENGINE WITH TIERED LOGIC

SDG: 44 –Subscription-Based Membership Engine with Tiered Logic

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INTRODUCTION

- Many online platforms such as streaming services, e-learning systems, and SaaS products use subscription models to provide services.
- Managing different membership levels manually becomes complex and error-prone.
- This project focuses on developing a Subscription-Based Membership Engine that supports multiple tiers like Basic, Premium, and Enterprise.
- The system automates subscription handling, payment validation, feature access control, and membership upgrades/downgrades.
- It ensures secure access control and efficient membership management.
- The project is relevant in real-world platforms like OTT services, online learning portals, and gym memberships.

PROBLEM STATEMENT

- Many businesses rely on manual or semi-automated systems to manage subscriptions.

- This causes issues such as:
  - ➢ Incorrect feature access
  - ➢ Poor tracking of subscription validity
  - ➢ Difficulty in managing upgrades/downgrades
  - ➢ Payment and renewal errors.

- There is a need for a secure, scalable, and automated system that manages tier-based membership logic efficiently.

OBJECTIVE(S)

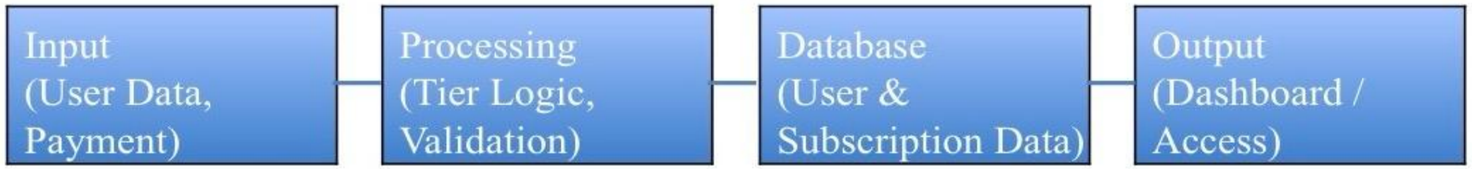
1. To design and develop a subscription management system with tier-based access control.

2. To implement automated renewal, upgrade, and downgrade logic for memberships.

PROPOSED SYSTEM

1. Users can register and choose subscription plans (Basic, Premium, Enterprise).
2. System validates payments and activates the subscription.
3. Access to features is controlled based on subscription tier.
4. Automatic expiration after subscription validity period.
5. Upgrade/Downgrade functionality with proper validation.
6. Admin panel to manage plans and users.
7. Email/Notification alerts before expiry.
8. Secure authentication and authorization mechanism.

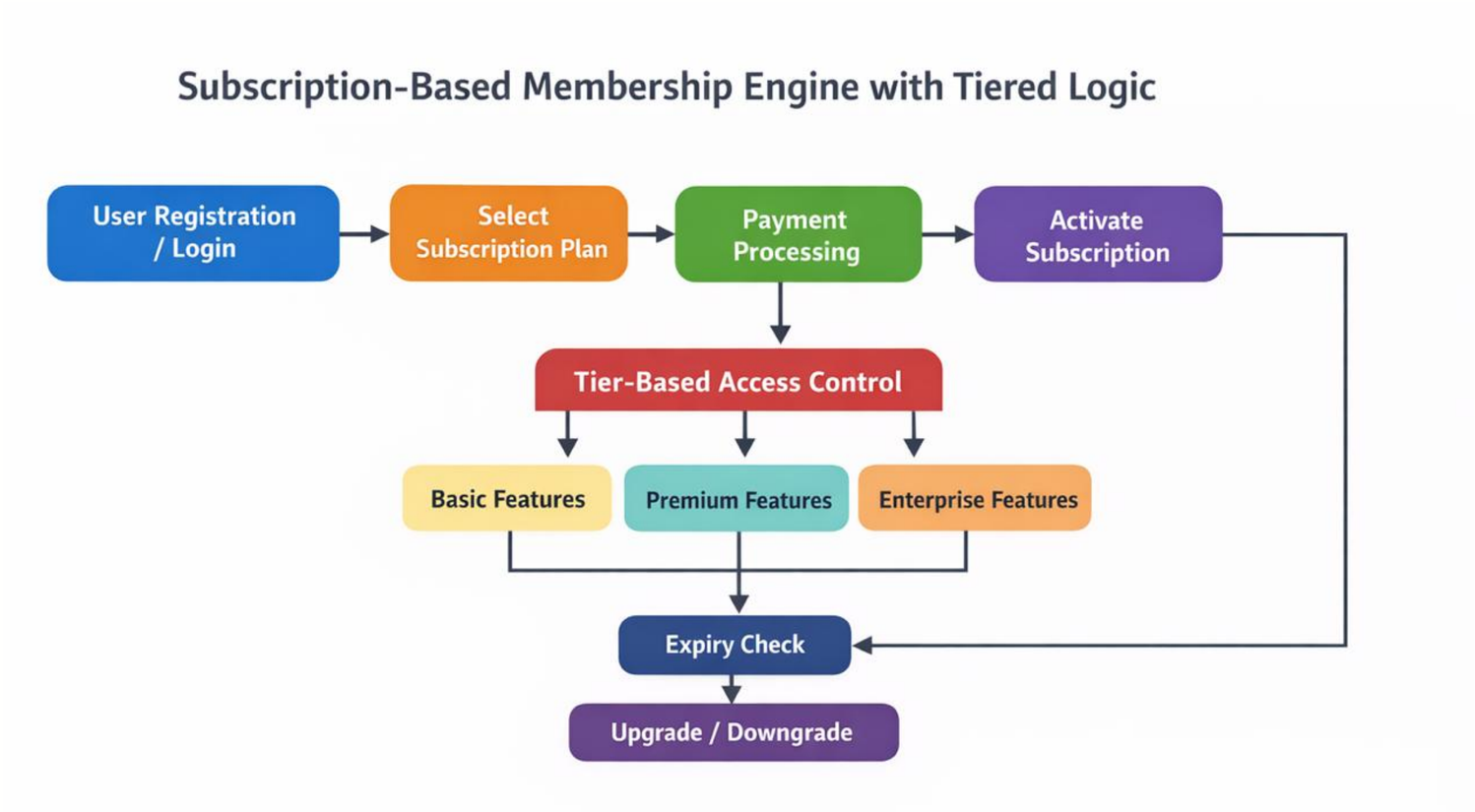
PROPOSED SYSTEM ARCHITECTURE



Explanation:

- Input: User registration, login, subscription selection, payment details
- Processing: Tier validation, access control logic, subscription status check
- Database: Stores user data, subscription plans, payment records
- Output: Dashboard access, subscription status display, restricted/allowed features

WORKFLOW



Dataflow:

- User registers/logs in
- Selects subscription tier
- Makes payment
- System verifies payment
- Subscription activated
- User accesses features based on tier
- System checks expiry date regularly

TECHNOLOGIES USED

- Frontend: HTML, CSS, JavaScript
- Backend: Node Js
- Database: MySQL
- Tools: VS Code
- Hardware/Software Requirements:
  - Minimum 8GB RAM
  - JDK 17
  - MySQL Server
  - Web Browser

MODULE DESCRIPTION

- **Module 1: User Management**
  - Handles user registration, login, authentication, and profile management.

- **Module 2: Subscription Management**
  - Manages subscription plans, tier logic, activation, renewal, upgrade, and downgrade.

- **Module 3: Payment Processing**
  - Validates payment details and updates subscription status securely.

- **Module 4: Access Control Engine**
  - Restricts or allows features based on user subscription tier.

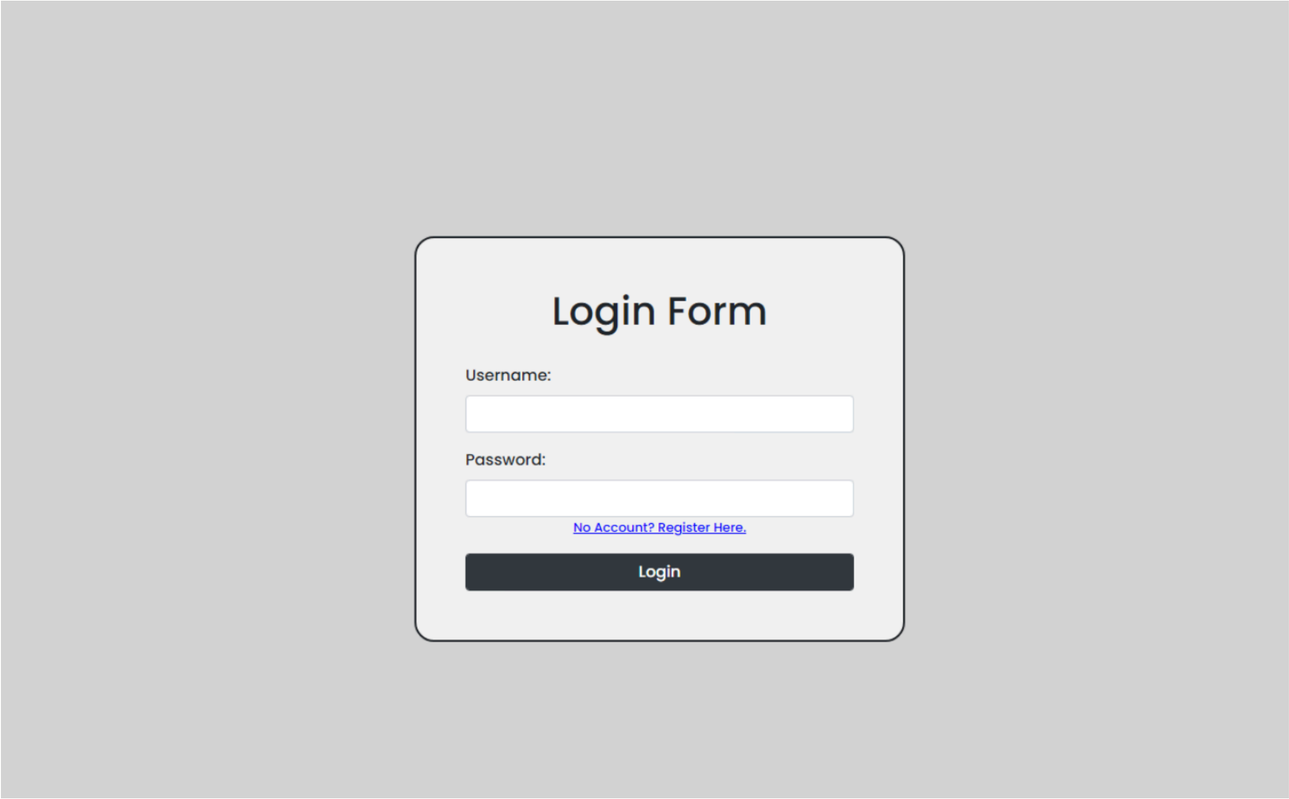
- **Module 5: Admin Dashboard**
  - Allows admin to manage users, plans, and monitor subscription analytics.

IMPLEMENTATION SCREENSHOTS

- **Home Page – Displays available subscription plans and system overview.**

IMPLEMENTATION SCREENSHOTS

➢ **Login Page – Allows users to securely log into the system.**

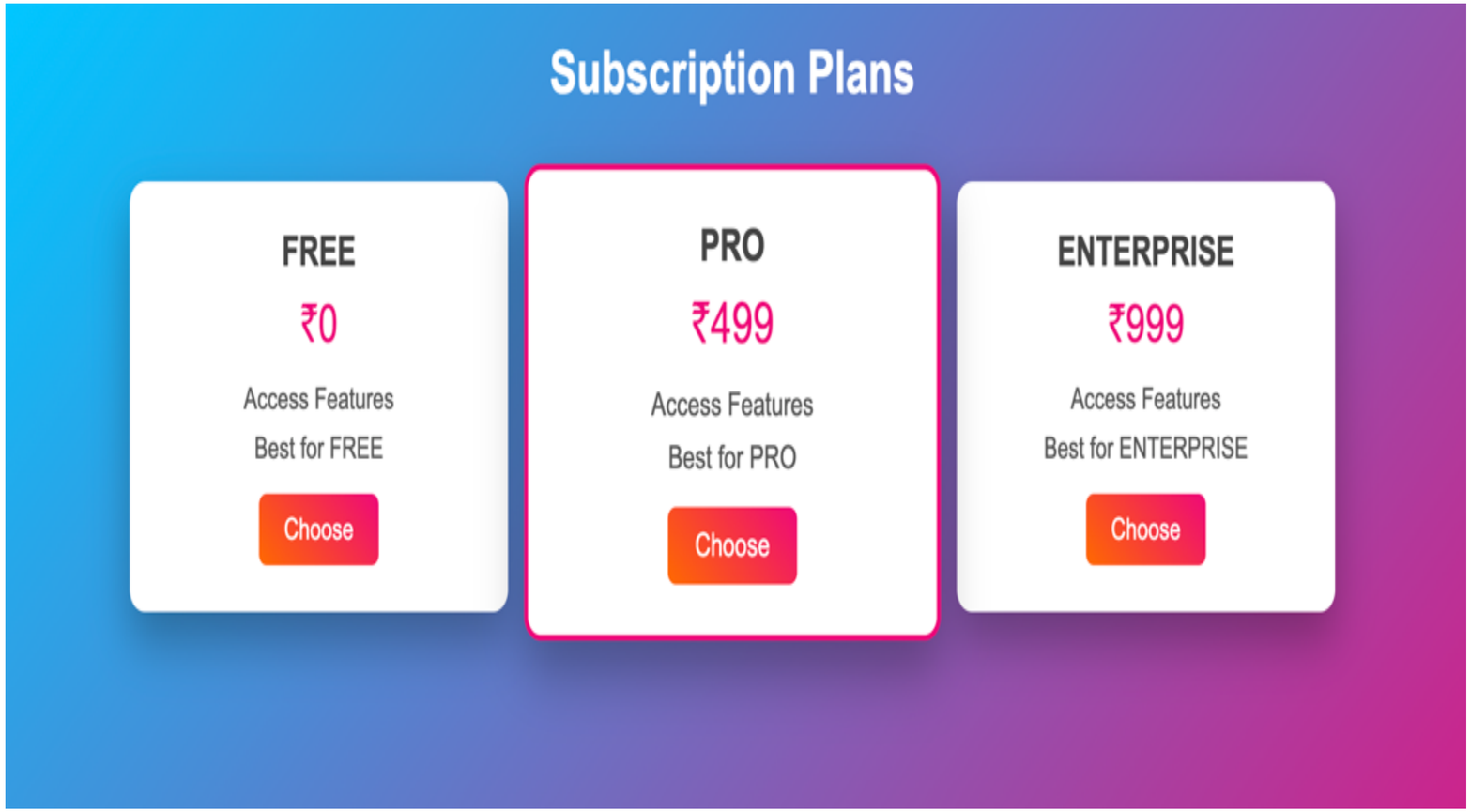


IMPLEMENTATION SCREENSHOTS

- **Main Dashboard – Shows active subscription details and accessible features.**

IMPLEMENTATION SCREENSHOTS

➢**Output Page – Displays subscription status, expiry date, and upgrade options.**



ADVANTAGES & LIMITATIONS

➢ **ADVANTAGES:**

- Automated subscription handling
- Secure tier-based access control
- Easy upgrade/downgrade system
- Reduces manual errors
- Scalable for large number of users

➢ **LIMITATIONS:**

- Requires secure payment gateway integration
- Internet connection required Initial setup cost

Thank you